

# Nethavath Praveen

Bachelors in Technology  
Electronics & Communication Engineering  
International Institute of Information Technology, Hyderabad

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## EDUCATION

|                                                                                                                                 |                                                               |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>International Institute of Information Technology (IIIT-H)</b><br><i>B.Tech in Electronics and Communication Engineering</i> | Hyderabad, India<br>CG: 6.53                                  |
| <b>Jawahar Navodaya Vidyalaya</b><br><i>12th Class (CBSE) — 80%</i>                                                             | Puducherry, Warangal, India<br><i>10th Class (CBSE) — 95%</i> |

## SKILLS SUMMARY

- **Languages:** Rust, JavaScript, SQL, Bash
- **Frameworks & APIs:** Node.js, FastAPI, React, HTML/CSS, REST APIs, GraphQL, WebSockets
- **Databases:** MongoDB, Redis, Neo4j
- **Cloud & DevOps:** AWS, Azure, GCP, Kubernetes, Harness, CI/CD, HashiCorp Vault, Observe, Webhooks, CRON
- **Concepts:** Microservices, Distributed Systems, Load Balancing, API Gateways, RAG, DSA, DBMS, OS, Computer Networks

## EXPERIENCE

- **Microland** Bangalore, India  
*Tech Intern — Platforms Team* May 2026 - Present
  - **Intelligeni IT Agent — Core CLI:** Designed and built from scratch a specialized **LLM-powered IT Service Engineer CLI in Rust** to **autonomously diagnose and remediate Azure Kubernetes (AKS)** infrastructure issues. Engineered a robust **3-layer execution permission system** (Hard Rails, dynamic command classification, and prompt approvals) to safely gate shell/destructive actions, and optimized remote troubleshooting by implementing persistent **SSH ControlMaster multiplexing** for low-latency command execution.
  - **Intelligeni IT Agent — Graph RAG:** Architected an **Enterprise Graph RAG platform** to trace alert propagation across **60+ microservices** for root-cause analysis. Integrated **Google Gemini 2.0 Flash** and a **Neo4j topology graph** to enable the IT agent to query and reason over complex microservice dependencies.
  - **Alert Correlation Architecture:** Re-engineered enterprise alert routing by transitioning a legacy **CRON** job to an **Observe** observability **webhook**, enabling real-time telemetry streaming. Designed and implemented an innovative **inter-instance relation logic** to correlate cross-region alerts and prevent ticket flooding.

## PROJECTS

- **Distributed Network File System** |  *C, TCP Sockets, POSIX Threads, LRU Cache, Trie*
  - Implemented a distributed file system from scratch supporting concurrent client-server communication using **TCP/IP sockets** and designed a multithreaded server architecture with **mutex synchronization**.
  - Integrated **LRU caching** and **Trie-based indexing** to reduce file lookup time by 35% and implemented file replication mechanisms for fault tolerance ensuring high availability.
- **Unix-like C Shell** |  *C, Linux Systems Programming, POSIX API*
  - Built a fully functional Unix-style shell supporting command piping, background job execution, and I/O redirection utilizing process lifecycle management via **fork()**, **exec()**, and **wait()** system calls.
  - Handled **SIGINT** and **SIGCHLD** signals effectively, preventing zombie processes through robust signal management and process group controls.
- **Codeforces Data Visualiser Website** |   *JavaScript, D3.js, React, CSS, Python, Flask*
  - Developed a **web platform** to analyze and visualize Codeforces user performance for **1,000+ users**, featuring interactive dashboards and responsive UI based on React and D3.js.
  - Integrated Codeforces API with a Python-Flask backend to deliver **10+ real-time** visualizations (e.g., rating trends, problem tags, contest history) with **<200 ms latency**, enhancing user insight and engagement.
- **Social Media Database Engine** |  *Python, MySQL*
  - Designed and implemented a normalized relational database schema for a social media platform, handling users, posts, hierarchical comments, and follower networks.
  - Developed a Python interface using **PyMySQL** to manage transactions and execute complex SQL queries utilizing **JOINS** and **aggregations** for analytics.
- **CAPTCHA Breaker** |  *Python, PyTorch, OpenCV, CNN/LSTM*
  - Developed a CAPTCHA solver by combining a **CNN-LSTM** architecture with **CTC loss** and **OpenCV** for color-based text processing.
  - Created an automated data generator for 1000+ synthetic CAPTCHAs with configurable fonts, noise, and difficulty levels for robust model training.

## RELEVANT COURSEWORK

Data Structures · Algorithm Design · Operating Systems · Database Systems · Machine Learning · Software Systems · Computer Networks · Computer Architecture · Probability & Statistics

## HONORS AND AWARDS

- Ranked Top 50 in Google BigCode 2026 (India); JEE Mains AIR 3137, JEE Advanced AIR 4894, FFE Scholar
- Successfully completed Zscaler Certifications (Sep 2025): Fundamentals of Cybersecurity (EDU-102) and Introduction to Networking for Cyber Professionals
- Served as Teaching Assistant for Probability and Random Processes (2025), Real Analysis (2025) at IIIT Hyderabad
- Served as NSS Co-ordinator (Spring 2025 – Present), Felicity Fest Marketing Team Member